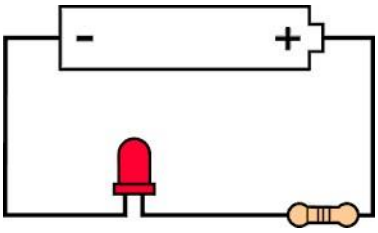
**Appendix D: Testing Procedure for Light Sensor & Competition Light Source Specifications**

Testing Procedure for Light Sensor: testing the light sensing apparatus (payload delivery system) and the process can be achieved using regular RED LED. A link is provided with RGB LED's that can be used to test the red sensing on your color-sensing unit. It should be noted that an appropriate resistor should be used to limit the current through the common ground on these LED's based on your input voltage (470 ohm works for 12V input, less for 5V)

| **Alternative Light Sensor Testing Equipment** | | | | | |
|---|---|---|---|---|---|
| Red LED | Red LED | Jameco | 1 | $0.25 | $0.25 |
| 9 Volt Battery | Energizer MAX 9 Volt | Amazon | 1 | $6.25 | $6.25 |
| 470 Ohm Resistor | 470 Ohm, 1/4 Watt Resistor | Amazon | 1 | $1.76 | $1.76 |
| | | | | **Total Cost** | **$8.26** |




Competition Light Source Specifications: The table below is the list of parts used to construct a light sources located in the payload delivery tower**. It is not necessary to build your own light source tower** to test the payload delivery system. Above mentioned testing procedures is applicable to test the payload delivery system.

| **UGV delivery-light source Components** | | | | | |
|---|---|---|---|---|---|
| LED Ring | NeoPixel Ring - 16 x 5050 RGB | Adafruit | 1 | $9.95 | $9.95 |
| Microcontroller | Adafruit Pro Trinket - 5V 16Mhz | Adafruit | 1 | $9.95 | $9.95 |
| LED Module | M625L3 –625nm Mounted LED | Thor Labs | 1 | $216.30 | $216.30 |
| | | | | **Total Cost** | $236.20 |

Note: The construction of the light source must follow the equipment and the company recommended guide lines.